WEEK 1 · LESSON 2 OF 24 · HANDOUT

A Short History of AI

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=fa8k8IQ1_X0 — open the lesson online to watch inside the course.

🎓 **Go deeper:** Harvard CS50x 2024: Artificial Intelligence — full lecture — https://www.youtube.com/watch?v=6X58aP7yXC4

Your goal in this lesson: Describe the main periods in the history of AI, explain why progress came in waves, and say what changed after 2012.

Why history matters

AI is not new. Knowing its history helps you judge today's claims. Again and again, people expected human-level AI "within 20 years" — and were wrong. Again and again, quiet research later became huge products.

The timeline

Year	Event	Why it matters
1950	Alan Turing publishes *Computing Machinery and Intelligence* and proposes the "imitation game" (the Turing Test)	Asked seriously: can machines think?
1956	The Dartmouth workshop; John McCarthy uses the term **"artificial intelligence"**	AI becomes a research field
1958	Frank Rosenblatt builds the **perceptron**, an early neural network	First machine that learned from examples
1970s	First **"AI winter"** — funding falls after promises are not met	Hype has a cost
1980s	**Expert systems** — big rule-based programs used in business	Symbolic AI goes commercial
late 1980s–1990s	Second AI winter; expert systems prove costly to maintain	Hand-written rules do not scale
1997	IBM's **Deep Blue** beats world chess champion Garry Kasparov	Search plus fast hardware wins
2012	A deep neural network (**AlexNet**) wins the ImageNet image-recognition contest by a wide margin	The **deep-learning era** begins
2016	DeepMind's **AlphaGo** beats Go champion Lee Sedol	Learning plus search beats humans in a far harder game
2017	Google researchers publish *Attention Is All You Need* — the **transformer**	The design behind today's language models
2022	**ChatGPT** is released and reaches millions of users within weeks	AI becomes an everyday tool

Why progress came in waves

Each big jump needed three things at the same time:

1. **Data** — the internet and smartphones created huge collections of text, images and sound.
2. **Computing power** — graphics chips (**GPUs**), first made for video games, turned out to be perfect for training neural networks.
3. **Better methods** — new ideas such as deep networks and transformers.

When one of the three was missing, progress slowed and funding fell — an AI winter. When all three came together after 2012, progress became very fast.

AI in Africa and Rwanda

Africa is not only a user of AI. Some examples:

- **Zipline** has delivered blood and medical supplies by autonomous drone in Rwanda since 2016, using software to plan routes and flights.
- Rwanda's Cabinet approved a **National AI Policy in 2023**, with goals for skills, data, infrastructure and responsible use.
- Kigali hosts the **Centre for the Fourth Industrial Revolution (C4IR) Rwanda** and has hosted international AI meetings, including the Global AI Summit on Africa in 2025.
- Researchers across the continent build tools for African languages such as Kinyarwanda, Swahili and Yoruba, which big global models often handle poorly.

Key terms

- **AI winter** — a period when interest and funding in AI fell sharply.
- **Expert system** — a program that stores expert knowledge as many rules.
- **GPU** — a graphics processing unit; it does many small calculations at once, which makes training fast.
- **Transformer** — the neural network design behind modern language models (Lesson 17).

Remember this: AI grew in waves. The current boom comes from data, computing power and better methods arriving together — and history warns us to be careful with hype.

Name: _____ Date: _____

Activity

Task 1

Pick one event from the timeline.

Task 2

Write a short paragraph (5–6 sentences): what happened, which of the three ingredients (data, computing, methods) made it possible, and what it changed.

Task 3

Find one recent AI headline. Write one sentence on whether it sounds like careful progress or hype, and why.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. Who proposed the 'imitation game', later called the Turing Test?
   A. John McCarthy
   B. Garry Kasparov
   C. Frank Rosenblatt
   D. Alan Turing

2. What happened at the 1956 Dartmouth workshop?

A. ChatGPT was released
B. The first GPU was built
C. The term 'artificial intelligence' was used and AI became a research field
D. Deep Blue beat Kasparov

3. What is an 'AI winter'?

A. A period when interest and funding in AI fell sharply
B. A cold-weather robot
C. The season when models are trained
D. A type of neural network

4. Which event is usually seen as the start of the deep-learning era?

A. The Dartmouth workshop
B. Deep Blue in 1997
C. Expert systems in the 1980s
D. AlexNet winning ImageNet in 2012

5. Which three ingredients drove the recent boom in AI?

A. Phones, apps and games only
B. Data, computing power and better methods
C. Robots, satellites and cables
D. Laws, schools and money

6. Which 2017 paper introduced the transformer?

A. Attention Is All You Need
B. The Perceptron
C. Computing Machinery and Intelligence
D. Deep Blue Report

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